

module5_lab2

October 4, 2026

```
[1]: import pandas as pd
df=pd.read_csv("Downloads/custdata.tsv", sep="\t")
```

```
[2]: df.head()
```

```
[2]:
```

	custid	sex	is.employed	income	marital.stat	health.ins	\
0	2068	F	NaN	11300	Married	True	
1	2073	F	NaN	0	Married	True	
2	2848	M	True	4500	Never Married	False	
3	5641	M	True	20000	Never Married	False	
4	6369	F	True	12000	Never Married	True	

	housing.type	recent.move	num.vehicles	age	state.of.res
0	Homeowner free and clear	False	2.0	49.0	Michigan
1	Rented	True	3.0	40.0	Florida
2	Rented	True	3.0	22.0	Georgia
3	Occupied with no rent	False	0.0	22.0	New Mexico
4	Rented	True	1.0	31.0	Florida

```
[3]: df.shape
```

```
[3]: (1000, 11)
```

```
[4]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   custid          1000 non-null  int64
1   sex             1000 non-null  str
2   is.employed     672 non-null   object
3   income          1000 non-null  int64
4   marital.stat    1000 non-null  str
5   health.ins      1000 non-null  bool
6   housing.type    944 non-null   str
7   recent.move     944 non-null   object
8   num.vehicles     944 non-null   float64
```

```
9   age          1000 non-null   float64
10  state.of.res  1000 non-null   str
dtypes: bool(1), float64(2), int64(2), object(2), str(4)
memory usage: 115.8+ KB
```

```
[5]: df.describe()
```

```
[5]:
```

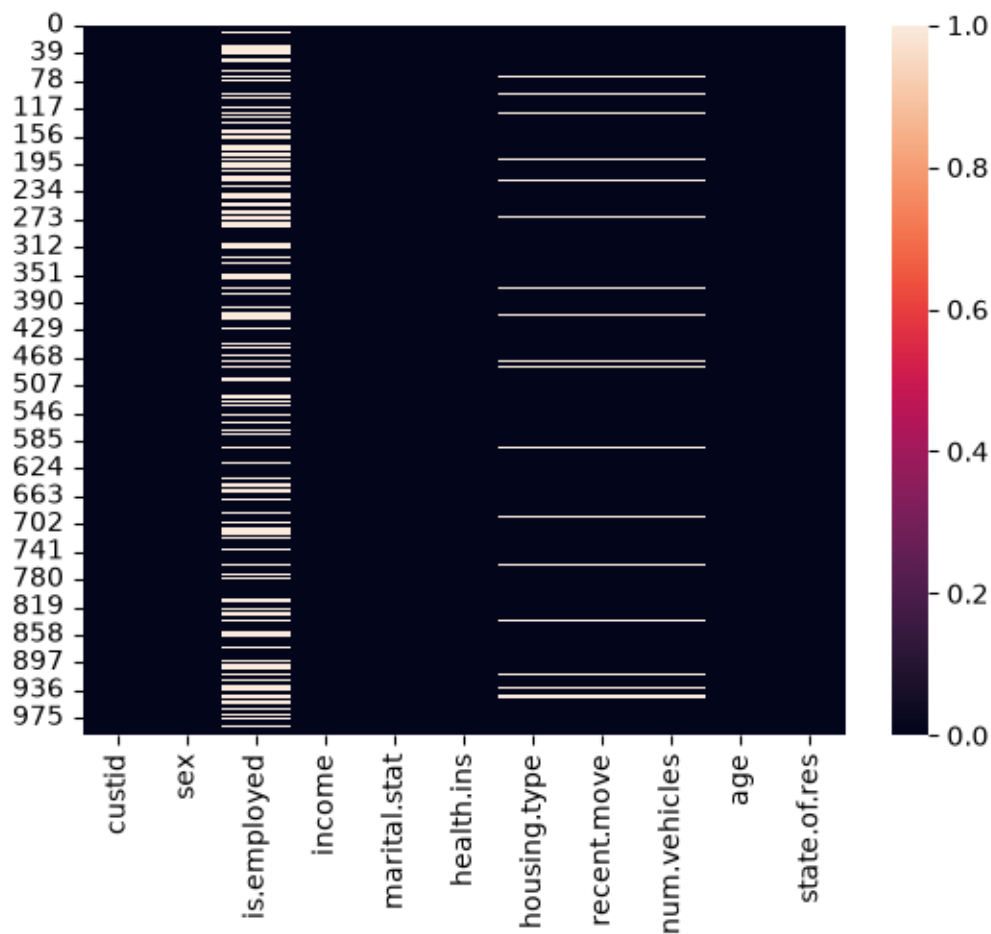
	custid	income	num.vehicles	age
count	1.000000e+03	1000.000000	944.000000	1000.000000
mean	6.984997e+05	53504.771000	1.916314	51.699815
std	4.135083e+05	65478.065729	1.101618	18.863433
min	2.068000e+03	-8700.000000	0.000000	0.000000
25%	3.456668e+05	14600.000000	1.000000	38.000000
50%	6.934030e+05	35000.000000	2.000000	50.000000
75%	1.044606e+06	67000.000000	2.000000	64.000000
max	1.414286e+06	615000.000000	6.000000	146.680197

```
[6]: df.isnull().sum()
```

```
[6]: custid          0
sex              0
is.employed     328
income          0
marital.stat    0
health.ins      0
housing.type    56
recent.move     56
num.vehicles     56
age             0
state.of.res    0
dtype: int64
```

```
[7]: import seaborn as sns
import matplotlib.pyplot as plt
sns.heatmap(df.isnull())
```

```
[7]: <Axes: >
```

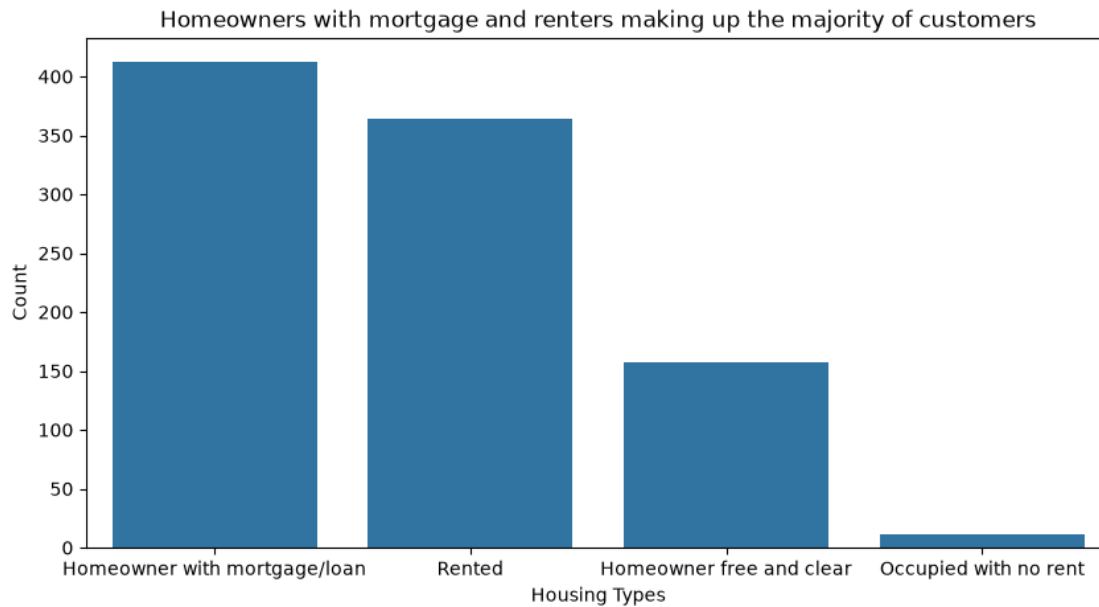


```
[8]: housing_clean=df.dropna(subset=["housing.type"])
```

```
[9]: housing_clean.isnull().sum()
```

```
[9]: custid      0
      sex        0
      is.employed 280
      income      0
      marital.stat 0
      health.ins   0
      housing.type 0
      recent.move  0
      num.vehicles  0
      age         0
      state.of.res 0
      dtype: int64
```

```
[10]: housing_counts = (housing_clean["housing.type"].value_counts().reset_index())
housing_counts.columns = ["housing_type", "count"]
plt.figure(figsize=(10,5))
sns.barplot(data=housing_counts,x="housing_type",y="count")
plt.title("Homeowners with mortgage and renters making up the majority of
↳customers")
plt.xlabel("Housing Types")
plt.ylabel("Count")
plt.show()
```



```
[11]: #markdown cell
#removed 56 rows of na data for housing type
```